

Cambridge (CIE) IGCSE Additional Maths



Your notes

Differentiation

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Introduction to Differentiation

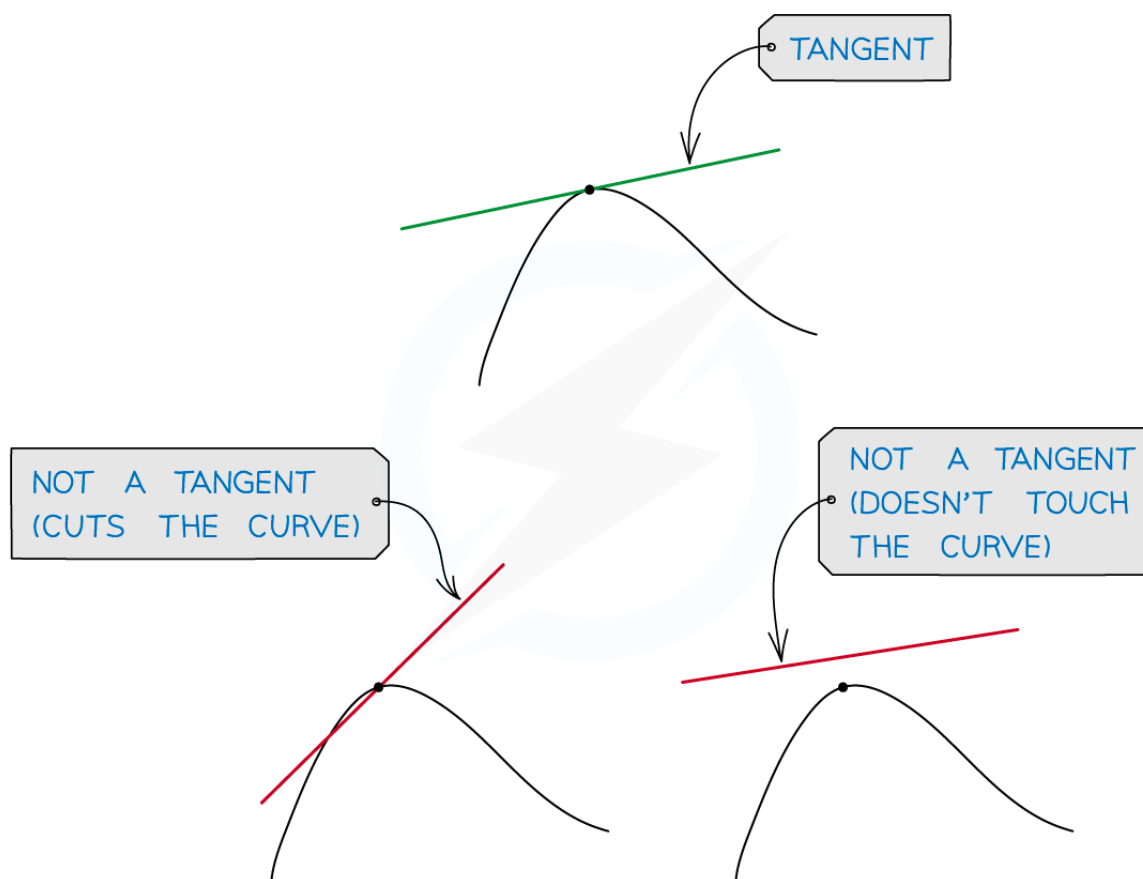


Your notes

Definition of Gradient

What is the gradient of a curve?

- At a given point the **gradient** of a curve is defined as the gradient of the **tangent** to the curve at that point
- A **tangent** to a curve is a line that just **touches** the curve at one point but doesn't cut the curve at that point

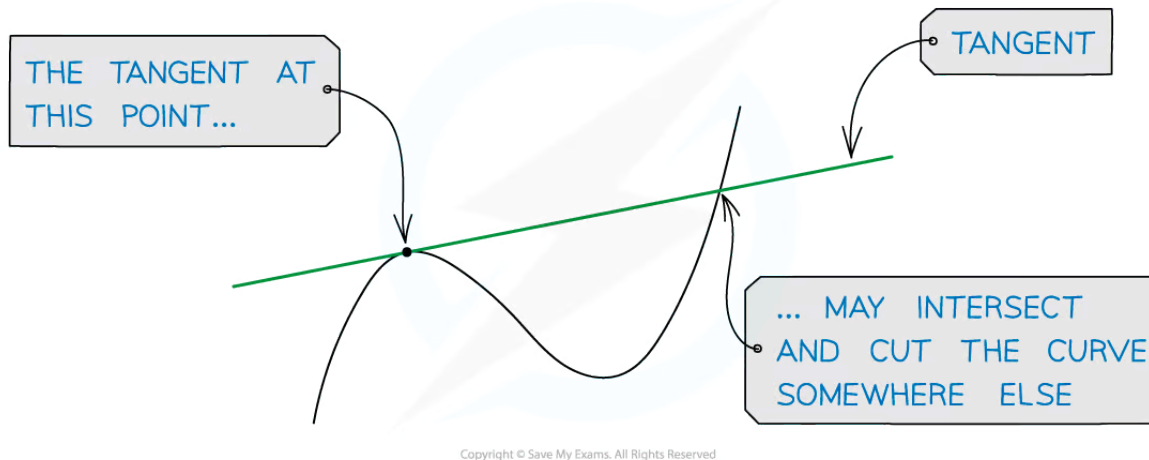


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- A tangent *may* cut the curve somewhere else on the curve



Your notes



- It is only possible to draw **one** tangent to a curve at any given point
- Note that unlike the gradient of a straight line, the gradient of a curve is constantly changing



Examiner Tips and Tricks

- If a question asks for the "rate of change of ..." then it is asking for the "gradient"

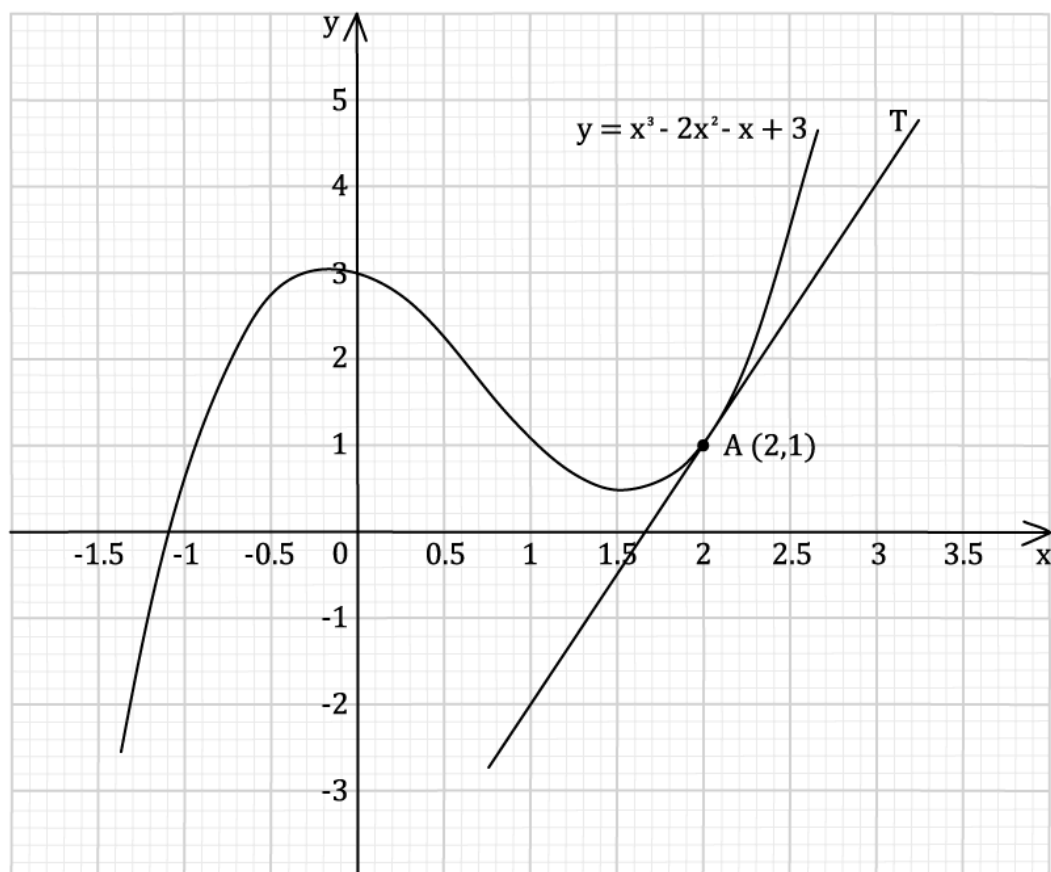


Worked Example

The diagram shows the curve with equation $y = x^3 - 2x^2 - x + 3$. The tangent, T , to the curve at the point $A(2, 1)$ is also shown.



Your notes

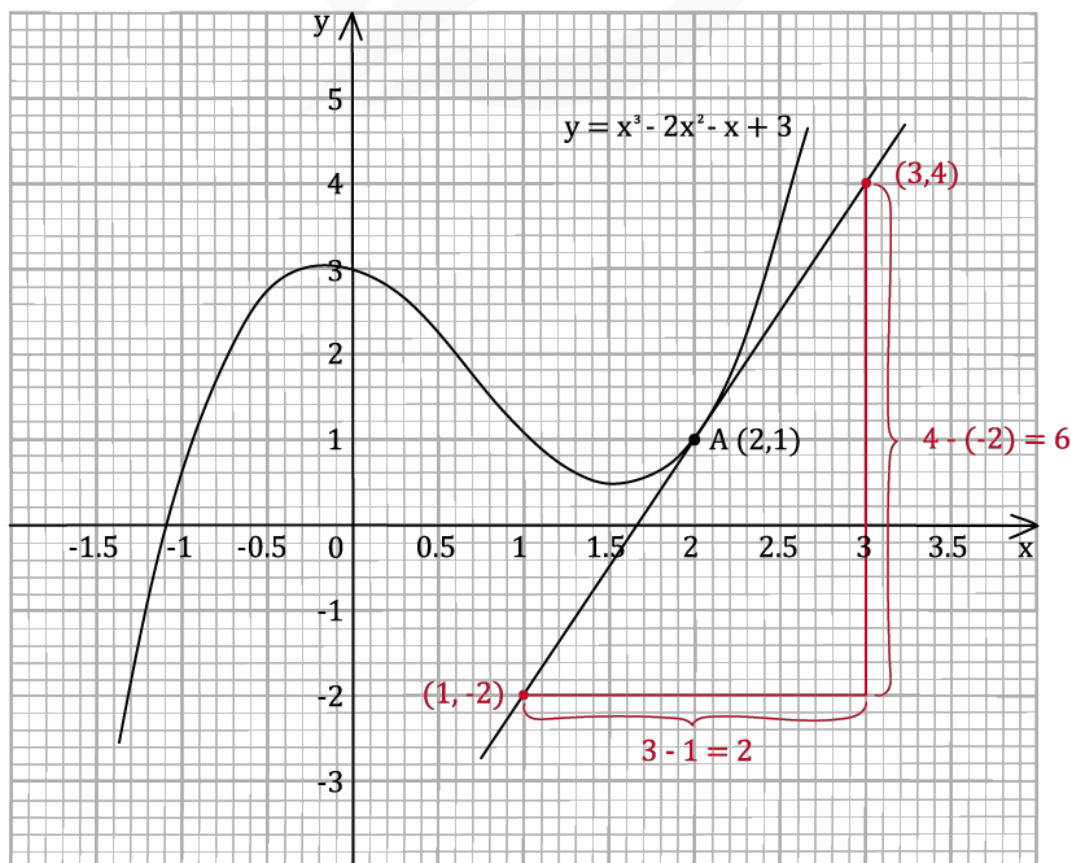


Using the diagram, calculate the gradient of the curve at A .

The gradient of the curve at the point A is the same as the gradient of the tangent T . Calculate the gradient of the line.



Your notes



$$\frac{4 - (-2)}{3 - 1}$$

The gradient is 3

Definition of Derivatives

What is a derivative?

- **Calculus** is about **rates of change**
 - the way a car's position on a road changes is its speed (velocity)
 - the way the car's speed changes is its acceleration
- The **gradient** (rate of change) of a (non-linear) **function** varies with x



Your notes

- The **derivative** of a function is a function that relates the **gradient** to the value of x
 - For example, the derivative of $y = x^2$ is $2x$
 - This means that when $x = 1$, the gradient of $y = x^2$ is $2(1) = 2$
 - And when $x = 5$, the gradient of $y = x^2$ is $2(5) = 10$
- The derivative is also called the **gradient function**



Worked Example

The derivative of $y = x^3 - 2x^2 - x + 3$ is $3x^2 - 4x - 1$.

Use the derivative to find the gradient of $y = x^3 - 2x^2 - x + 3$ at the point $A(2, 1)$.

Substitute $x = 2$ into the derivative, $3x^2 - 4x - 1$

$$3(2)^2 - 4(2) - 1 = 12 - 8 - 1$$

gradient = 3

Note that the answer is the same as in the method above

Differentiating Powers of x

What is differentiation?

- **Differentiation** is the process of finding an expression for the **derivative (gradient function)** from the equation of a curve

- The equation of the curve is written $y = \dots$ and the gradient function is written $\frac{dy}{dx} = \dots$

How do I differentiate powers of x ?

- Powers of x are **differentiated** according to the following formula:

- If $y = ax^n$ then $\frac{dy}{dx} = anx^{n-1}$



Your notes

- e.g. If $y = 4x^3$ then $\frac{dy}{dx} = 4 \times 3 \times x^{3-1} = 12x^2$
- you "bring down the power" then "subtract one from the power"
- Don't forget these **two** special cases:
 - If $y = ax$ then $\frac{dy}{dx} = a$
 - e.g. If $y = 6x$ then $\frac{dy}{dx} = 6$
 - If $y = a$ then $\frac{dy}{dx} = 0$
 - e.g. If $y = 5$ then $\frac{dy}{dx} = 0$
- These allow you to differentiate **linear terms** in x and **constants**
- Functions involving **fractions** with **denominators** in terms of x will need to be rewritten as **negative powers** of x first

- e.g. If $y = \frac{4}{x}$ then rewrite as $y = 4x^{-1}$ and differentiate

How do I differentiate sums and differences of powers of x ?

- The formulae for differentiating powers of x work for a **sum** or **difference** of **powers** of x
 - e.g. If $y = 5x^4 + 3x^{-2} + 4$ then $\frac{dy}{dx} = 5 \times 4x^{4-1} + 3 \times (-2)x^{-2-1} + 0$
 - $\frac{dy}{dx} = 20x^3 - 6x^{-3}$
 - This is sometimes referred to differentiating 'term-by-term'
- **Products** and **quotients (divisions)** **cannot** be differentiated in this way so they need **expanding/simplifying** first
 - e.g. If $y = (2x - 3)(x^2 - 4)$ then expand to $y = 2x^3 - 3x^2 - 8x + 12$ which is a **sum/difference** of powers of x and can then be differentiated



Your notes

What can I do with derivatives (gradient functions)?

- The **derivative** can be used to find the **gradient** of a **function** at any point
 - The gradient of a function at a point is equal to the **gradient** of the **tangent** to the curve at that point
 - A question may refer to the gradient of the tangent



Examiner Tips and Tricks

- Don't try to do too many steps in your head; write the expression in a format that you can differentiate before you actually differentiate it
 - e.g. $y = \frac{1}{x^4} + \frac{2}{x^3}$ can be rewritten as $y = x^{-4} + 2x^{-3}$ which is then far easier to differentiate



Worked Example

Find the derivative of

(a) $y = 5x^3 + 2x + \frac{3}{x^2} + 8$

Rewrite the $\frac{3}{x^2}$ term

$$y = 5x^3 + 2x + 3x^{-2} + 8$$

Apply the rule for differentiating powers ($y = ax^n$, $\frac{dy}{dx} = anx^{n-1}$) and apply the special cases

for the terms $2x$ and 8 ($y = ax$, $\frac{dy}{dx} = a$ and $y = a$, $\frac{dy}{dx} = 0$)

$$\frac{dy}{dx} = 15x^2 + 2 - 6x^{-3}$$



Your notes

Unless a question specifies there is not usually a need to rewrite/simplify the answer

$$\frac{dy}{dx} = 15x^2 + 2 - \frac{6}{x^3}$$

(b) $y = (2x + 3)^2$

This is a product of two (equal) brackets so cannot be differentiated directly Expand the brackets to get an expression in powers of x Take time to get the expansion correct, writing stages out in full if necessary

$$\begin{aligned}y &= (2x + 3)(2x + 3) \\y &= 4x^2 + 6x + 6x + 9 \\y &= 4x^2 + 12x + 9\end{aligned}$$

Differentiate 'term-by-term', looking out for those special cases

$$\frac{dy}{dx} = 8x + 12$$

There is a factor of 4 but there is no demand to factorise the final answer in the question

$$\frac{dy}{dx} = 8x + 12$$

(c) $y = \frac{8x^6 - x^3}{2x^4}$

This is a quotient so cannot be differentiated directly Spot the single denominator which means we can split the fraction by the two terms on the numerator

$$y = \frac{8x^6}{2x^4} - \frac{x^3}{2x^4}$$

Simplify using the laws of indices



Your notes

$$y = 4x^{6-4} - \frac{1}{2}x^{3-4}$$

$$y = 4x^2 - \frac{1}{2}x^{-1}$$

Each term is now a power of x , so differentiate 'term-by-term'

$$\frac{dy}{dx} = 8x + \frac{1}{2}x^{-2}$$

There is demand to simplify or write the answer in a particular form

$$\frac{dy}{dx} = 8x + \frac{1}{2}x^{-2}$$



Your notes

Differentiating Special Functions

Differentiating Trig Functions

How do I differentiate trig functions?

- For calculus with trigonometric functions, angles must be measured in **radians**
- The derivative of $y = \sin x$ is $\frac{dy}{dx} = \cos x$
- The derivative of $y = \cos x$ is $\frac{dy}{dx} = -\sin x$
- The derivative of $y = \tan x$ is $\frac{dy}{dx} = \sec^2 x$
- The following two sets of relationships can be derived using the **chain rule**, but are useful to know!
- For the **linear** function $ax + b$, where a and b are constants,
 - the derivative of $y = \sin(ax + b)$ is $\frac{dy}{dx} = a \cos(ax + b)$
 - the derivative of $y = \cos(ax + b)$ is $\frac{dy}{dx} = -a \sin(ax + b)$
 - the derivative of $y = \tan(ax + b)$ is $\frac{dy}{dx} = a \sec^2(ax + b)$
- For the **general** function $f(x)$,
 - the derivative of $y = \sin(f(x))$ is $\frac{dy}{dx} = f'(x) \cos(f(x))$
 - the derivative of $y = \cos(f(x))$ is $\frac{dy}{dx} = -f'(x) \sin(f(x))$
 - the derivative of $y = \tan(f(x))$ is $\frac{dy}{dx} = f'(x) \sec^2(f(x))$





Your notes

Examiner Tips and Tricks

- Remember that these rules only work in radians!



Worked Example

a) Find $f'(x)$ for the functions

i. $f(x) = \sin x$

ii. $f(x) = \cos 2x$

iii. $f(x) = 3\sin 4x - \cos(2x - 3)$

i.

$$f'(x) = \cos x$$

ii. Use the chain rule or remember that when $y = \cos(ax + b)$, then $\frac{dy}{dx} = -a\sin(ax + b)$

$$f'(x) = -2\sin(2x)$$

iii. Differentiate 'term by term'

$$f'(x) = 3(4\cos 4x) - (-2\sin(2x - 3))$$

$$f'(x) = 12\cos 4x + 2\sin(2x - 3)$$

b) Find the gradient of the tangent to the curve $y = \sin\left(2x + \frac{\pi}{6}\right)$ at the point where

$$x = \frac{\pi}{8}$$

Gradient of tangent is equal to gradient of curve. To find the gradient, differentiate...

$$\frac{dy}{dx} = 2\cos\left(2x + \frac{\pi}{6}\right)$$

... and substitute $x = \frac{\pi}{8}$ into the derivative



Your notes

$$\frac{dy}{dx} = 2\cos\left(2\left(\frac{\pi}{8}\right) + \frac{\pi}{6}\right)$$

Ensure that your calculator is in radians

$$= \frac{\sqrt{6} - \sqrt{2}}{2} = 0.517638090...$$

$$\text{Answer} = \frac{\sqrt{6} - \sqrt{2}}{2}, \text{ or } 0.518 \text{ to 3 significant figures}$$

Differentiating e^x & $\ln x$

How do I differentiate exponentials and logarithms?

- The derivative of $y = e^x$ is $\frac{dy}{dx} = e^x$ where $x \in \mathbb{R}$
- The derivative of $y = \ln x$ is $\frac{dy}{dx} = \frac{1}{x}$ where $x > 0$
- In addition, the results below can all be found using the **chain rule** but are worthwhile knowing, to save time in an exam
- For the **linear** function $ax + b$, where a and b are constants,
 - the derivative of $y = e^{(ax+b)}$ is $\frac{dy}{dx} = ae^{(ax+b)}$
 - the derivative of $y = \ln(ax+b)$ is $\frac{dy}{dx} = \frac{a}{(ax+b)}$
 - in the special case $b = 0$, $\frac{dy}{dx} = \frac{1}{x}$ (a 's cancel)
- For the **general** function $f(x)$,
 - the derivative of $y = e^{f(x)}$ is $\frac{dy}{dx} = f'(x)e^{f(x)}$

- the derivative of $y = \ln(f(x))$ is $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$



Examiner Tips and Tricks

- Remember to avoid the common mistakes:
 - the derivative of $\ln kx$ with respect to x is $\frac{1}{x}$, NOT $\frac{k}{x}$!
 - the derivative of e^{kx} with respect to x is ke^{kx} , NOT kxe^{kx-1} !



Worked Example

A curve has the equation $y = e^{-3x} + 2\ln x$.

Find the gradient of the curve at the point where $x = 2$ giving your answer in the form $a + be^c$, where a , b and c are integers to be found.

Differentiate each term separately.

$$\frac{dy}{dx} = -3e^{-3x} + 2\left(\frac{1}{x}\right)$$

Substitute $x = 2$ into the derivative.

$$\frac{dy}{dx} = -3e^{-3(2)} + 2\left(\frac{1}{2}\right)$$

Rearrange to the required form.

$$\frac{dy}{dx} = -3e^{-6} + 1 = 1 - 3e^{-6}$$

Do not use your calculator to evaluate this as the question asks for the answer given in this form.

$$1 - 3e^{-6}$$



Your notes

Chain Rule

Chain Rule

What is the chain rule?

- The **chain rule** states if y is a function of u and u is a function of x then

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

- In **function notation** this could be written

$$y = f(g(x))$$

$$\frac{dy}{dx} = f'(g(x))g'(x)$$

How do I know when to use the chain rule?

- The chain rule is used when we are trying to differentiate **composite functions**
 - “function of a function”
 - these can be identified as the variable (usually x) does not ‘appear alone’
 - $\sin x$ – **not** a composite function, x ‘appears alone’
 - $\sin(3x + 2)$ is a **composite function**; x is tripled and has 2 added to it before the sine function is applied

How do I use the chain rule?

STEP 1

Identify the two functions

Rewrite y as a function of u ; $y = f(u)$

Write u as a function of x ; $u = g(x)$

STEP 2



Your notes

Differentiate y with respect to u to get $\frac{dy}{du}$ Differentiate u with respect to x to get $\frac{du}{dx}$

STEP 3

Obtain $\frac{dy}{dx}$ by applying the formula $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ and substitute u back in for $g(x)$

- In trickier problems **chain rule** may have to be applied **more than once**

How do I differentiate $(ax + b)^n$?

- For $n = 2$ you will most likely **expand the brackets** and differentiate each term separately
- If $n > 2$ this becomes **time-consuming** and if n is not a positive integer we need a different method completely
- The chain rule allows us to use **substitution** to differentiate any function in the form $y = (ax + b)^n$

- Let $u = ax + b$, then $y = u^n$

- Differentiate** both parts **separately**

$$\frac{du}{dx} = a \text{ and } \frac{dy}{du} = nu^{n-1}$$

- Put both parts into the **chain rule**

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = a \times nu^{n-1} = anu^{n-1}$$

- Substitute** $u = ax + b$ back into your answer

$$\frac{dy}{dx} = an(ax + b)^{n-1}$$

How do I differentiate $\sqrt{ax+b}$?

- The chain rule allows us to use **substitution** to differentiate any function in the form $y = \sqrt{ax + b}$

$$\text{Rewrite } \sqrt{ax + b} = (ax + b)^{\frac{1}{2}}$$

- Let $u = ax + b$, then $y = u^{\frac{1}{2}}$



Your notes

- **Differentiate** both parts **separately**

$$\frac{du}{dx} = a \text{ and } \frac{dy}{du} = \frac{1}{2} u^{-\frac{1}{2}}$$

- Put both parts into the **chain rule**

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = a \times \frac{1}{2} u^{-\frac{1}{2}} = \frac{a}{2} u^{-\frac{1}{2}}$$

- **Substitute** $u = ax + b$ back into your answer

$$\frac{dy}{dx} = \frac{a}{2} (ax + b)^{-\frac{1}{2}} = \frac{a}{2\sqrt{ax + b}}$$

- This method can be used for any **fractional power** of any linear or non-linear expression
 - Provided you know how to differentiate the non-linear expression

Are there any standard results for using chain rule?

- The following general results are particularly useful

- If $y = (f(x))^n$ then $\frac{dy}{dx} = nf'(x)f(x)^{n-1}$

- If $y = e^{f(x)}$ then $\frac{dy}{dx} = f'(x)e^{f(x)}$

- If $y = \ln(f(x))$ then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

- If $y = \sin(f(x))$ then $\frac{dy}{dx} = f'(x)\cos(f(x))$

- If $y = \cos(f(x))$ then $\frac{dy}{dx} = -f'(x)\sin(f(x))$

- If $y = \tan(f(x))$ then $\frac{dy}{dx} = f'(x)\sec^2(f(x))$



Examiner Tips and Tricks



Your notes

- You should aim to be able to spot and carry out the chain rule mentally (rather than use substitution)
 - every time you use it, say it to yourself in your head "differentiate the first function ignoring the second, then multiply by the derivative of the second function"



Worked Example

a) Find the derivative of $y = (x^2 - 5x + 7)^7$.

STEP 1 Identify the two functions and rewrite

$$y = u^7 \quad u = x^2 - 5x + 7$$

STEP 2 Find $\frac{dy}{du}$ and $\frac{du}{dx}$

$$\frac{dy}{du} = 7u^6 \quad \frac{du}{dx} = 2x - 5$$

STEP 3 Apply the chain rule, $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$$\frac{dy}{dx} = 7u^6(2x - 5)$$

Substitute u in terms of x back in

$$= 7(x^2 - 5x + 7)^6(2x - 5)$$

$$\frac{dy}{dx} = 7(2x - 5)(x^2 - 5x + 7)^6$$

b) Find the derivative of $y = \sin(e^{2x})$.

(In this solution, we will be applying the mental method discussed in the Exam Tip above)

"... differentiate $\sin \square$, ignore e^{2x} "

$$\frac{dy}{dx} = \cos(e^{2x}) \times \square$$

"... multiply by the derivative of e^{2x} ": differentiate e^{2x} using the result "if $y = e^{f(x)}$, then

$$\frac{dy}{dx} = f'(x)e^{f(x)}$$

$$\frac{dy}{dx} = \cos(e^{2x}) \times 2e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x} \cos(e^{2x})$$



Your notes



Your notes

Product Rule

Product Rule

What is the product rule?

- The product rule is a formula that allows you to differentiate a **product of two functions**
- If $y = u \times v$ where u and v are functions of x then the product rule is:

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

- In **function notation**, if $f(x) = g(x) \times h(x)$ then the product rule can be written as:

$$f'(x) = g(x)h'(x) + h(x)g'(x)$$

- The easiest way to remember the product rule is, for $y = u \times v$ where u and v are functions of x :

$$y' = uv' + vu'$$

How do I know when to use the product rule?

- The **product rule** is used when we are trying to **differentiate** the **product of two functions**
 - These can easily be confused with composite functions (see **chain rule**)
 - $\sin(\cos x)$ is a composite function, "sin of cos of x "
 - $\sin x \cos x$ is a product, "sin x times cos x "

How do I use the product rule?

- Make it clear what u , v , u' and v' are
 - arranging them in a square can help
 - opposite diagonals match up

STEP 1



Your notes

Identify the two functions, u and v

Differentiate both u and v with respect to x to find u' and v'

STEP 2

Obtain $\frac{dy}{dx}$ by applying the product rule formula $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$ Simplify the answer if straightforward to do so or if the question requires a particular form

e.g. $y = xe^x \leftarrow \boxed{u = x, v = e^x}$

$$\frac{dy}{dx} = (x)(e^x) + (e^x)(1)$$

Diagram showing the product rule application for $y = xe^x$. Arrows point from u to (x) , from $\frac{dv}{dx}$ to (e^x) , from v to (e^x) , and from $\frac{du}{dx}$ to (1) .

$$\frac{dy}{dx} = xe^x + e^x$$

$$f(x) = x^3(x^2 + 7)$$

$$f'(x) = (x^3)(2x) + (x^2 + 7)(3x^2) \leftarrow \boxed{g(x) = x^3, h(x) = x^2 + 7}$$

Diagram showing the product rule application for $f(x) = x^3(x^2 + 7)$. Arrows point from $g(x)$ to (x^3) , from $h'(x)$ to $(2x)$, from $h(x)$ to $(x^2 + 7)$, and from $g'(x)$ to $(3x^2)$.

$$f'(x) = 5x^4 + 21x^2$$

COMPARE

$$\begin{aligned} f(x) &= x^3(x^2 + 7) \\ f(x) &= x^5 + 7x^3 \\ f'(x) &= 5x^4 + 21x^2 \end{aligned}$$

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Your notes

 Examiner Tips and Tricks

- The product rule formula is **not** on the list of formulas page – make sure you know it
- Don't confuse the **product** of two functions with a **composite function**:
 - The **product** of two functions is two functions **multiplied** together
 - A **composite** function is a **function of a function**

e.g. $f(x) = 3x^2$ $g(x) = x - 3$

$$f(x)g(x) = 3x^2(x - 3) = 3x^3 - 9x^2$$

PRODUCT

$$fg(x) = f(g(x)) = 3(x - 3)^2 = 3x^2 - 18x + 27$$

COMPOSITE
FUNCTION

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- To differentiate composite functions you need to use the **chain rule**



Worked Example



Your notes

? Differentiate with respect to x :

a) $(5\sin x - 7\cos x)e^{3x+2}$

b) $5x\ln 5x$

a) $u = 5\sin x - 7\cos x$

$v = e^{3x+2}$

$\frac{du}{dx} = 5\cos x + 7\sin x$

$\frac{dv}{dx} = 3e^{3x+2}$

DON'T FORGET MINUS
SIGN WHEN YOU
DIFFERENTIATE $\cos$

USE CHAIN RULE

$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

$\frac{dy}{dx} = (5\sin x - 7\cos x)(3e^{3x+2}) + (e^{3x+2})(5\cos x + 7\sin x)$

$\frac{dy}{dx} = (22\sin x - 16\cos x)e^{3x+2}$

b) $u = 5x$

$v = \ln 5x$

$\frac{du}{dx} = 5$

$\frac{dv}{dx} = \frac{1}{x}$

BE CAREFUL WITH THIS ONE!
BECAUSE OF THE CHAIN RULE
THE DERIVATIVE OF $\ln(ax)$
IS $\frac{1}{x}$ FOR ANY CONSTANT a

$\frac{dy}{dx} = (5x)\left(\frac{1}{x}\right) + (\ln 5x)(5)$

$\frac{dy}{dx} = 5 + 5\ln 5x$



Your notes

Quotient Rule

Quotient Rule

What is the quotient rule?

- The quotient rule is a formula that allows you to differentiate a **quotient of two functions**
 - i.e. one function divided by another
- If $y = \frac{u}{v}$ where u and v are functions of x then the quotient rule is:

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

- In **function notation**, if $f(x) = \frac{g(x)}{h(x)}$ then the quotient rule can be written as:

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$$

- As with the product rule, '**dash notation**' may be used to make remembering it easier

$$y = \frac{u}{v}$$

$$y' = \frac{vu' - uv'}{v^2}$$

- Final answers should match the notation used throughout the question

How do I know when to use the quotient rule?

- The **quotient rule** is used when trying to differentiate a fraction where **both** the **numerator** and **denominator** are **functions** of x

- if the **numerator** is a **constant**, **negative powers** can be used
- if the **denominator** is a **constant**, treat it as a **factor** of the expression\



Your notes

How do I use the quotient rule?

- Make it clear what u , v , u' and v' are
 - arranging them in a square can help
 - opposite diagonals match up (like they do for product rule)

STEP 1

Identify the two functions, u and v

Differentiate both u and v with respect to x to find u' and v'

STEP 2

Obtain $\frac{dy}{dx}$ by applying the quotient rule formula $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ Be careful using the formula –

because of the **minus sign** in the **numerator**, the **order** of the functions is important Simplify the answer if straightforward or if the question requires a particular form



Your notes

e.g. $y = \frac{x^2+1}{x-3} \leftarrow \boxed{u = x^2+1, v = x-3}$

$$\frac{dy}{dx} = \frac{(x-3)(2x) - (x^2+1)(1)}{(x-3)^2}$$

$$\frac{dy}{dx} = \frac{x^2 - 6x - 1}{(x-3)^2}$$

$f(x) = \frac{\sin x}{\cos x} \leftarrow \boxed{g(x) = \sin x, h(x) = \cos x}$

$$f'(x) = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$f'(x) = \sec^2 x \leftarrow \boxed{\frac{1}{\cos x} = \sec x}$$

$$\boxed{\frac{\sin x}{\cos x} = \tan x \text{ SO THIS IS THE DERIVATIVE OF } \tan x}$$

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Your notes

Examiner Tips and Tricks

- The quotient rule formula is **not** on the list of formulas page – you have to memorise it
 - however if you do forget it in an exam, you could rewrite $y = \frac{u}{v}$ as $y = u \times v^{-1}$ and then use the **product rule**
 - (the quotient rule is really doing exactly this)
- Be careful using the formula – because of the minus sign in the numerator the order of the functions is important!



Worked Example



Your notes

? Differentiate $f(x) = (\cos 2x)(x^2 + 2x + 1)^{-1}$
with respect to x .

$$f(x) = \frac{\cos 2x}{x^2 + 2x + 1}$$

REWRITE AS $f(x) = \frac{g(x)}{h(x)}$

$$g(x) = \cos 2x$$

$$h(x) = x^2 + 2x + 1$$

$$g'(x) = -2\sin 2x$$

$$h'(x) = 2x + 2$$

CHAIN RULE – AND REMEMBER MINUS
SIGN WHEN DIFFERENTIATING $\cos$

$$f'(x) = \frac{(x^2 + 2x + 1)(-2\sin 2x) - (\cos 2x)(2x + 2)}{(x^2 + 2x + 1)^2}$$

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$$

$$f'(x) = \frac{-2\sin 2x(x^2 + 2x + 1) - 2\cos 2x(x + 1)}{(x^2 + 2x + 1)^2}$$

$$= \frac{-2\sin 2x(x + 1)^2 - 2\cos 2x(x + 1)}{(x + 1)^4}$$

$$x^2 + 2x + 1 = (x + 1)^2$$

$$f'(x) = \frac{-2(\cos 2x + (x + 1)\sin 2x)}{(x + 1)^3}$$

SIMPLIFY IT



Your notes

Applications of Differentiation

Finding Gradients

How do I use the derivative to find the gradient of a curve?

- The **gradient of a curve** at a point is the **gradient of the tangent** to the curve at that point
- To find the gradient of a curve, at any point on the curve

- differentiate** to find $\frac{dy}{dx}$ (unless $\frac{dy}{dx}$ is already known)

- substitute the **x-coordinate** of the point into the derivative $\frac{dy}{dx}$ and evaluate

e.g. FIND THE GRADIENT OF THE CURVE

$$y = x - \frac{1}{x} \quad \text{AT THE POINT } (2, \frac{3}{2})$$

$$y = x - x^{-1}$$

$$\frac{dy}{dx} = 1 - (-1)x^{-1-1} = 1 + x^{-2} = 1 + \frac{1}{x^2}$$

FIND
DERIVATIVE

WHEN $x = 2$,

$$\frac{dy}{dx} = 1 + \frac{1}{2^2} = \frac{5}{4}$$

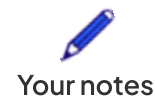
PUT x-COORDINATE
INTO DERIVATIVE

THE GRADIENT AT $(2, \frac{3}{2})$ IS $\frac{5}{4}$

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How do I find the approximate change in y as x increases?

- gradient = $\frac{\text{change in } y}{\text{change in } x}$ so, for **small** changes you can write



- change in $y = \text{gradient} \times \text{change in } x$
- For example, if the gradient of $y = x - \frac{1}{x}$ at $x = 2$ is $\frac{5}{4}$
 - what is the approximate change in y as x increases from 2 to $2 + h$, where h is small?
 - change in $y = \text{gradient} \times \text{change in } x = \frac{5}{4}h$



Examiner Tips and Tricks

- Read the question carefully; sometimes you are given $\frac{dy}{dx}$ and so don't need to differentiate initially - don't just automatically differentiate the first thing you see!
- The following mean the same thing:
 - "Find the gradient of the curve at $x = 2$ "
 - "Find the gradient of the tangent at $x = 2$ "
 - the tangent gradient = curve gradient at that point
 - "Find the **rate of change** of y with respect to x at $x = 2$ "



Worked Example

A curve has the equation $y = \frac{4}{3}x^3 + 3x - 8$.

(a) Find the gradient of the curve when $x = 2$.

y is already in a form that can be differentiated

$$\frac{dy}{dx} = 4x^2 + 3$$

Substitute $x = 2$ into $\frac{dy}{dx}$



Your notes

$$\frac{dy}{dx} = 4 \times 2^2 + 3 = 19$$

The gradient of the curve at $x = 2$ is 19

(b) Work out the possible values of x for which the rate of change of y with respect to x is 4.

"Rate of change" is another way of describing the derivative

$$\begin{aligned}\frac{dy}{dx} &= 4 \\ 4x^2 + 3 &= 4\end{aligned}$$

Solve this equation to find x

Note that it is quadratic equation so it could have up to two solutions

The question refers to 'values' implying there is (or could be) more than one value for x

$$\begin{aligned}4x^2 &= 1 \\ x^2 &= \frac{1}{4} \\ x &= \pm \sqrt{\frac{1}{4}} \\ x &= \pm \frac{1}{2}\end{aligned}$$

The possible values of x , that give a rate of change of 4, are $x = \frac{1}{2}$ and $x = -\frac{1}{2}$

Increasing & Decreasing Functions

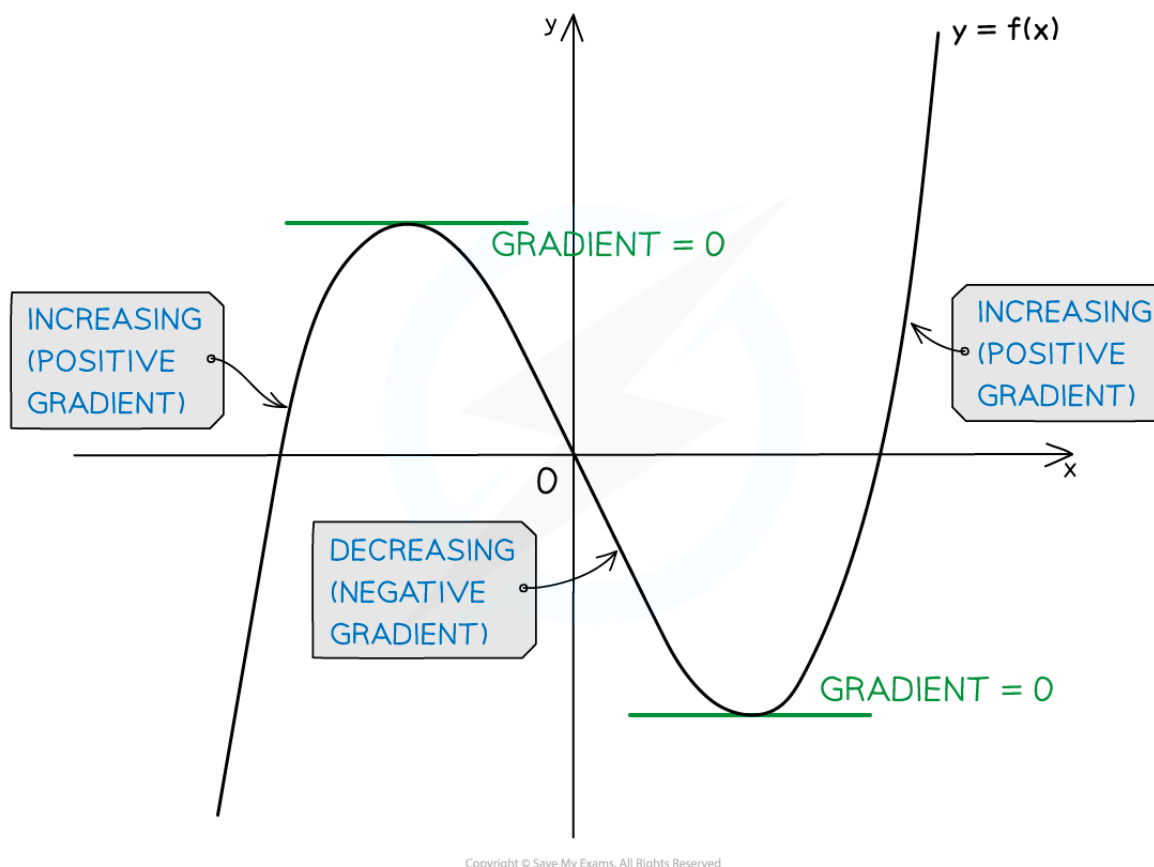
What are increasing and decreasing functions?

- A function is **increasing** when $\frac{dy}{dx} > 0$ (the gradient is positive)
 - This means graph of a function goes **up** as x increases
- A function is **decreasing** when $\frac{dy}{dx} < 0$ (the gradient is negative)

- This means graph of a function goes **down** as **x** increases



Your notes



How do I find where functions are increasing or decreasing?

- To identify the **intervals** on which a function is increasing or decreasing

STEP 1

Find the derivative $f'(x)$

STEP 2

Solve the inequalities

$f'(x) > 0$ (for increasing intervals) and/or



Your notes

$f'(x) < 0$ (for decreasing intervals)

- Most functions are a combination of **increasing**, **decreasing** and **stationary**
 - a range of values of x (**interval**) is given where a function satisfies each condition
 - e.g. The function $f(x) = x^2$ has **derivative** $f'(x) = 2x$ so
 - $f(x)$ is **decreasing** for $x < 0$
 - $f(x)$ is **stationary** at $x = 0$
 - $f(x)$ is **increasing** for $x > 0$
- To identify the intervals (the range of x values) for which a curve is increasing or decreasing you need to:

1. Find the derivative $\frac{dy}{dx}$

2. Solve the inequalities $\frac{dy}{dx} > 0$ (for increasing intervals) or $\frac{dy}{dx} < 0$ (for decreasing intervals)



Examiner Tips and Tricks

- In an exam, if you need to show a function is increasing or decreasing you can use either strict ($<$, $>$) or non-strict ($\leq$, $\geq$) inequalities
- You will get the marks either way in this course



Worked Example

For what values of x is $y = 2x^3 - 3x^2 + 5$ a decreasing function?

The function is decreasing when its gradient $\left(\frac{dy}{dx}\right)$ is less than 0.

Find the derivative of the function by differentiating.



Your notes

$$\frac{dy}{dx} = 6x^2 - 6x$$

Solve the inequality $\frac{dy}{dx} < 0$ to find the set of values where the gradient is negative.

$$6x^2 - 6x < 0$$

Factorise.

$$6x(x - 1) < 0$$

The solutions to $\frac{dy}{dx} = 0$ are $x = 0$ and $x = 1$. Find the correct way around for the inequalities

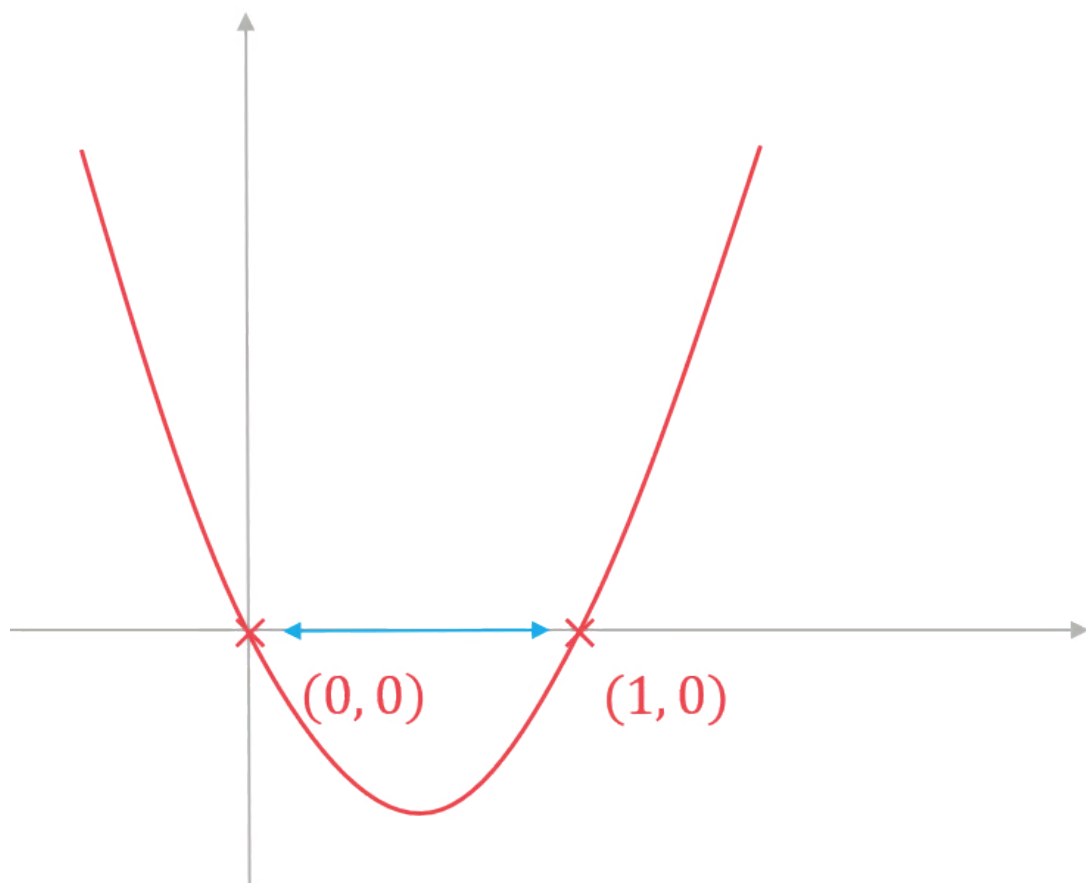
by considering the graph of $\frac{dy}{dx}$. The graph is a positive quadratic, so the function is negative

between the values of 0 and 1 (where it is below the x -axis).

Considering a sketch of the graph of the gradient function may help you see this.



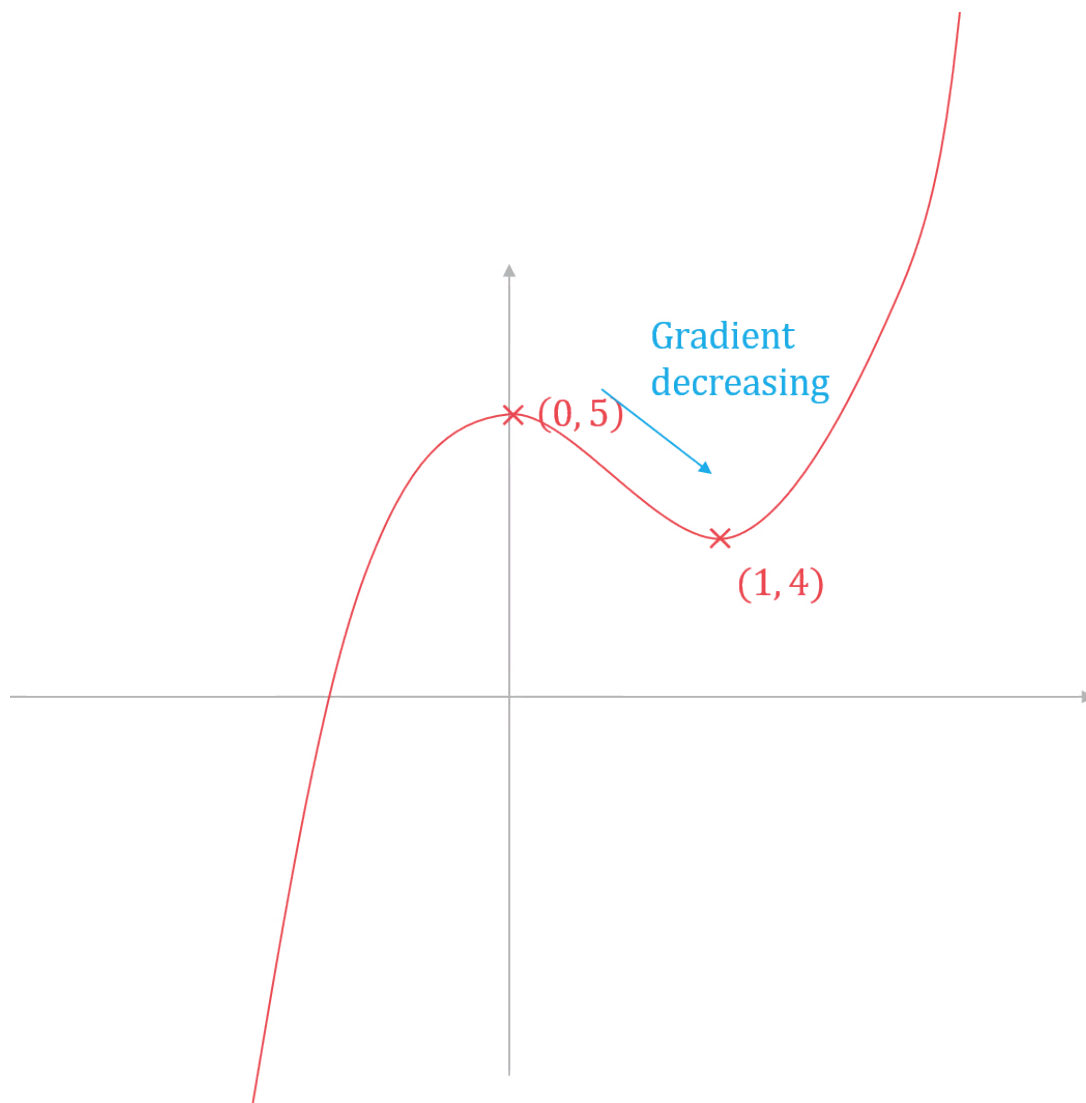
Your notes



You can check your answers by considering a sketch of the original function, it should be decreasing at the point where $0 < x < 1$.



Your notes



$$0 < x < 1$$

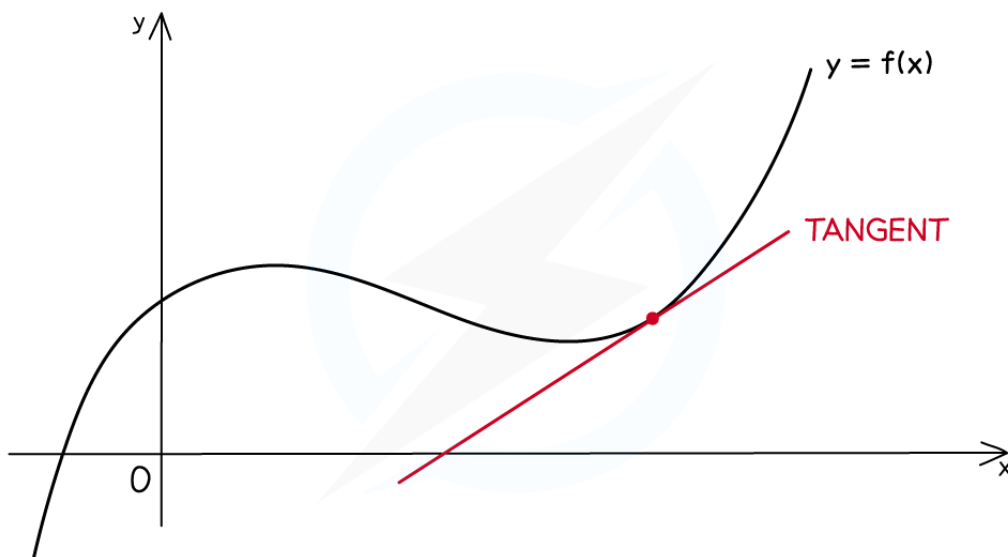
Tangents & Normals

What is a tangent?

- At any point on the graph of a (non-linear) **function**, the **tangent** is the straight line that **touches** the graph at a point **without crossing** through it
- Its **gradient** is given by the **derivative function**



Your notes



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How do I find the equation of a tangent?

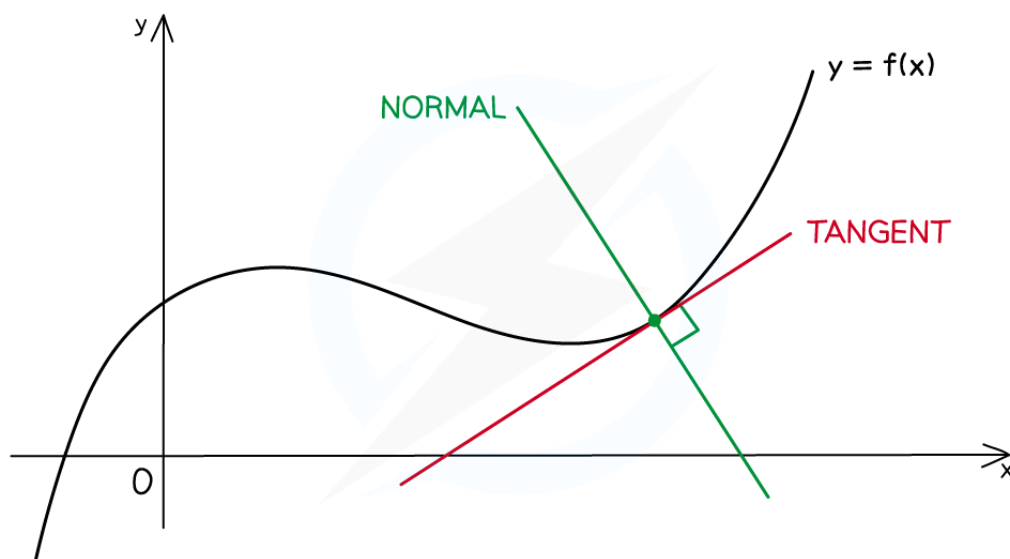
- To find the **equation of a straight line**, a **point** and the **gradient** are needed
- The **gradient**, m , of the **tangent** to the function $y = f(x)$ at (x_1, y_1) is $f'(x_1)$
 - You can find this by differentiating the function, and then substituting the x -coordinate of the point into the derivative
- Therefore find the **equation** of the **tangent** to the function $y = f(x)$ at the point (x_1, y_1) by substituting the gradient, $f'(x_1)$, and point (x_1, y_1) into $y - y_1 = m(x - x_1)$, giving:
 - $y - y_1 = f'(x_1)(x - x_1)$
- (You could also substitute into $y = mx + c$)

What is a normal?

- At any point on the graph of a (non-linear) function, the **normal** is the straight line that passes through that point and is **perpendicular** to the **tangent**



Your notes



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How do I find the equation of a normal?

- The gradients of two perpendicular lines are **negative reciprocals**
 - This means that if m_1 is the gradient of the first line and m_2 is the gradient of a line perpendicular to the first line, then $m_2 = \frac{-1}{m_1}$
 - Rearranging the formula above, $m_1 \times m_2 = -1$ is a useful way to test whether two lines are perpendicular

- Therefore **gradient** of the **normal** to the function $y = f(x)$ at (x_1, y_1) is $\frac{-1}{f'(x_1)}$

- Find the **equation** of the **normal** to the function $y = f(x)$ at the point (x_1, y_1) by using

$$y - y_1 = \frac{-1}{f'(x_1)}(x - x_1)$$

- (or $y = \frac{-1}{f'(x_1)}x + c$)



Your notes



Examiner Tips and Tricks

- To be successful in this topic, first make sure you are confident with finding the equation of a straight line!



Worked Example

The function $f(x)$ is defined by

$$f(x) = 2x^4 + \frac{3}{x^2} \quad x \neq 0$$

a) Find an equation for the tangent to the curve $y = f(x)$ at the point where $x = 1$, giving your answer in the form $y = mx + c$.

First find the derivative $f'(x)$ by differentiating

Start by rewriting $f(x)$ as powers of x

$$f(x) = 2x^4 + 3x^{-2}$$

Now differentiate

$$f'(x) = 8x^3 - 6x^{-3}$$

Now substitute $x = 1$ into $f'(x)$ to find the gradient of the tangent

$$\begin{aligned} f'(1) &= 8(1)^3 - 6(1)^{-3} \\ &= 8 - 6 \\ &= 2 \end{aligned}$$

We also need the y -coordinate, so substitute $x = 1$ into $f(x)$ also

$$\begin{aligned} f(1) &= 2(1)^4 + 3(1)^{-2} \\ &= 2 + 3 \\ &= 5 \end{aligned}$$



Your notes

Now we can substitute the point (1, 5) and the gradient, 2, into $y - y_1 = m(x - x_1)$

$$y - 5 = 2(x - 1)$$

Note that we are asked for the final answer in the form $y = mx + c$, so rearrange to this form

$$y - 5 = 2x - 2$$

$$y = 2x + 3$$

b) Find an equation for the normal at the point where $x = 1$, giving your answer in the form $ax + by + d = 0$, where a , b and d are integers.

We already have the gradient of the tangent; the gradient of the normal is $-\frac{1}{m}$ where m is the gradient of the tangent

$$\text{gradient of normal} = \frac{-1}{2}$$

Substitute the point (1, 5) from part (a) and the gradient of the normal into $y - y_1 = m(x - x_1)$

$$y - 5 = \left(\frac{-1}{2}\right)(x - 1)$$

And rearrange into the form required

$$y - 5 = -\frac{1}{2}x + \frac{1}{2}$$

$$\frac{1}{2}x + y - 5\frac{1}{2} = 0$$

Note that a , b and d must be integers so multiply by 2 to clear the fractions

$$x + 2y - 11 = 0$$



Your notes

Second Order Derivatives

Second Order Derivatives

What is the second order derivative of a function?

- If you **differentiate the derivative** of a function (i.e. differentiate the function a second time) you get the **second order derivative** of the function
 - The second order derivative can be referred to simply as the second derivative
- We can write the second derivative as $\frac{d^2y}{dx^2}$
- Note the position of the powers of 2
 - differentiating twice (so **d²**) with respect to **x** twice (so **x²**)
- A **first derivative** is the **rate of change** of a function (the gradient)
 - a **second order derivative** is the **rate of change of the rate of change** of a function
 - i.e. the rate of change of the function's gradient
 - A positive second derivative means the gradient is increasing
 - For instance in a u-shape, the gradient is changing from negative to positive
 - A negative second derivative means the gradient is decreasing
 - For instance in an n-shape, the gradient is changing from positive to negative
- Second order derivatives can be used to test whether a point is a minimum or maximum
- To find a second derivative, you simply **differentiate twice!**
 - It is important to write down your working with the correct notation, so you know what each expression means
 - For example
 - $y = 5x^3 + 10x^2$
 - $\frac{dy}{dx} = 15x^2 + 20x$

$$\frac{d^2y}{dx^2} = 30x + 20$$



Your notes



Examiner Tips and Tricks

- Even if you think you can find the second derivative in your head and write it down, make sure you write down the first derivative as well
- If you make a mistake, you will most likely get marks for finding the first derivative



Worked Example

Work out $\frac{d^2y}{dx^2}$ when

(a) $y = x^5 - 2x^3 + 7x^2 + 9x - 18$

Find the first derivative of the function first by considering each term in turn.

$$\frac{dy}{dx} = 5x^4 - 6x^2 + 14x + 9$$

Find the second derivative, $\frac{d^2y}{dx^2}$ by differentiating each term in the first derivative.

$$\frac{d^2y}{dx^2} = 20x^3 - 12x + 14$$

(b) $y = \frac{3x + 7}{x^4}$

To find the first derivative of the function, begin by separating the terms in the fraction.



Your notes

$$y = \frac{3x}{x^4} + \frac{7}{x^4}$$

Rewrite each term using index notation so that they are in a form that can be differentiated.

$$y = 3x(x^{-4}) + 7x^{-4} = 3x^{-3} + 7x^{-4}$$

Find the first derivative of the function first by differentiating each term in turn.

$$\frac{dy}{dx} = -9x^{-4} - 28x^{-5}$$

Find the second derivative, $\frac{d^2y}{dx^2}$ by differentiating each term in the first derivative.

$$\frac{d^2y}{dx^2} = -9(-4)x^{-5} - 28(-5)x^{-6} = 36x^{-5} + 140x^{-6}$$

You can turn the second derivative back into the same format as the original function by rewriting as a fraction.

$$\frac{d^2y}{dx^2} = \frac{36}{x^5} + \frac{140}{x^6} = \frac{36x + 140}{x^6}$$

$$\frac{d^2y}{dx^2} = \frac{36x + 140}{x^6}$$

Stationary Points & Turning Points

What are stationary points?

- A **stationary point** is any point on a curve where the **gradient** is **zero**
- To find stationary points of a curve

Step 1 Find the first derivative $\frac{dy}{dx}$

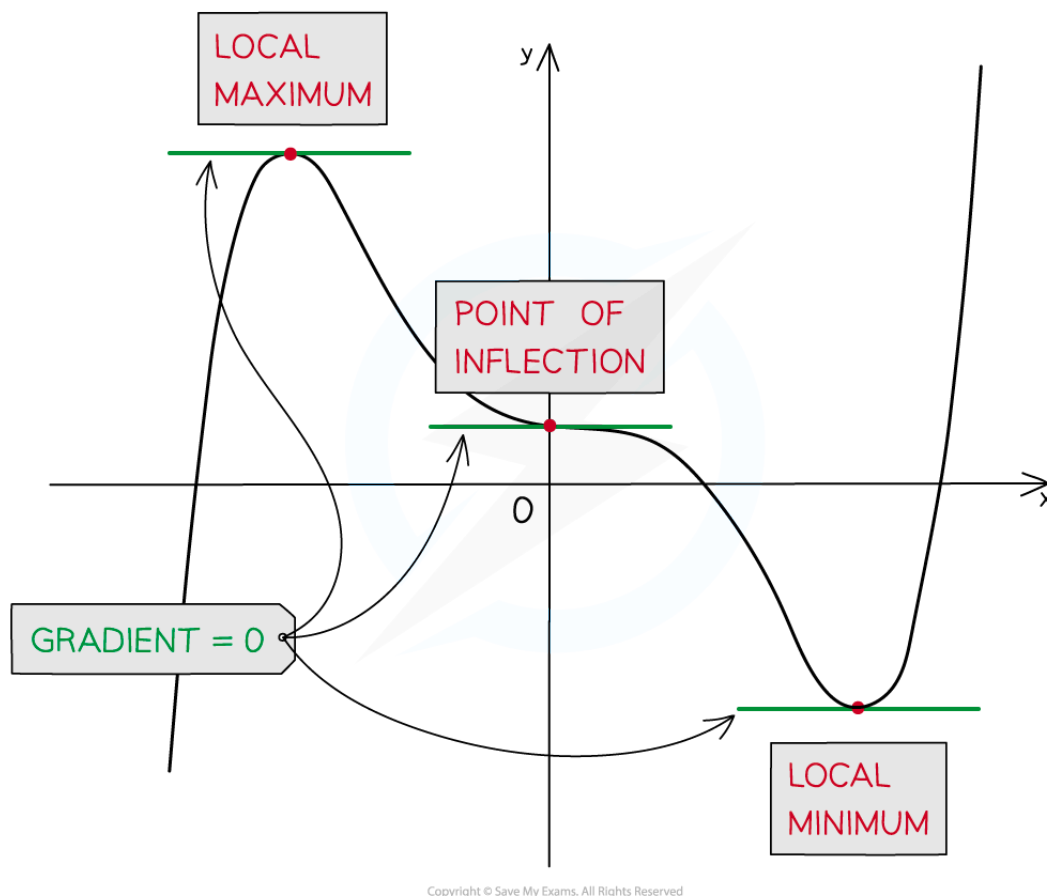
Step 2 Solve $\frac{dy}{dx} = 0$ to find the **X**-coordinates of any stationary points

Step 3 Substitute those **X**-coordinates into the equation of the curve to find the corresponding **y**-coordinates

- A stationary point may be either a **local minimum**, a **local maximum**, or a **point of inflection**



Your notes

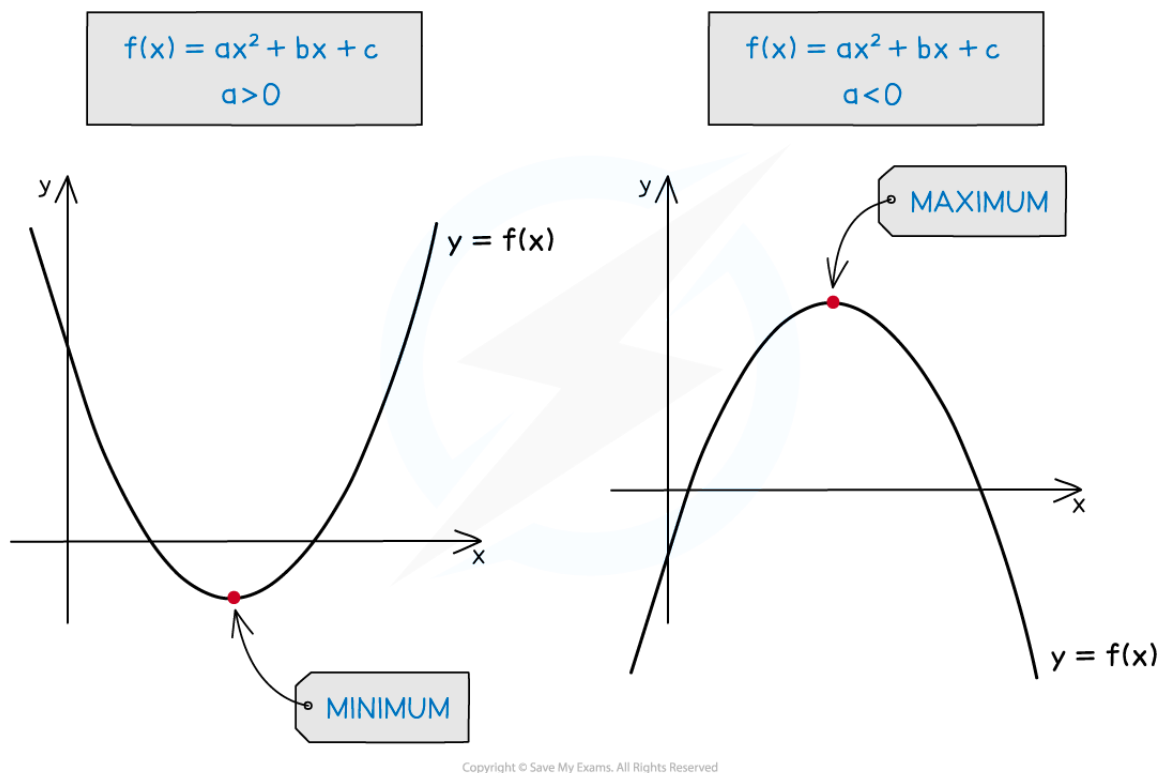


Stationary points on quadratics

- The graph of a **quadratic** function only has a **single stationary point**
- For a **positive** quadratic this is the **minimum**; for a **negative** quadratic it is the **maximum**
 - No need to talk about 'local' here, as it is the overall minimum/maximum for the whole curve



Your notes



- The **y** value/coordinate of the stationary point is therefore the minimum or maximum value of the quadratic function
- For quadratics especially, minimum and maximum points are often referred to as **turning points**



Worked Example

Find the stationary point of

$$y = x^2 - 2x$$

and explain why it will be a minimum point.

Find the derivative.

$$\frac{dy}{dx} = 2x - 2$$



Your notes

Solve $\frac{dy}{dx} = 0$

$$2x - 2 = 0$$

$$2x = 2$$

$$x = 1$$

Find the corresponding y -coordinate.

$$x = 1, y = (1)^2 - 2(1) = -1$$

Write the answer as a coordinate.

The stationary point is (1, -1)

To explain why it is a minimum point, consider the shape of the quadratic.

(1, -1) is a minimum point because it is a positive quadratic.

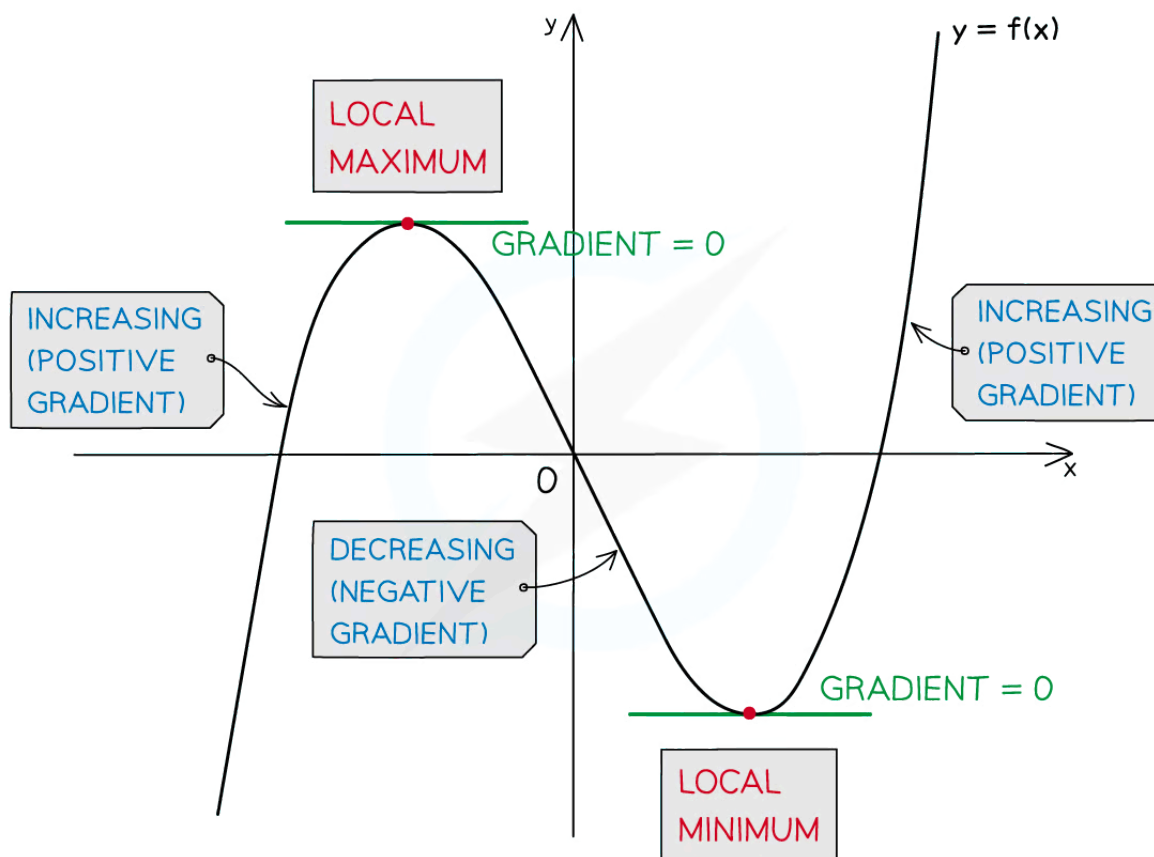
Testing for Local Minimum & Maximum Points

How do I determine the nature of stationary points on a curve?

- For a graph there are two ways to determine the nature of its stationary points
- Method A**
 - Compare the signs of the **first derivative**, $\frac{dy}{dx}$, (positive or negative) a little bit **to either side of the stationary point**
 - e.g. if the stationary point is at $x=2$ then you could find the gradient at $x=1.9$ and $x=2.1$
 - Compare the **signs** (positive or negative) of the derivatives on the left and right of the stationary point
 - If the derivatives are **negative** on the **left** and **positive** on the **right**, the point is a **local minimum** (a **u-shape**)
 - If the derivatives are **positive** on the **left** and **negative** on the **right**, the point is a **local maximum** (an **n-shape**)



Your notes



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Method B

- Look at the sign of the **second derivative** (positive or negative) **at the stationary point**

- Find the second derivative $\frac{d^2 y}{dx^2}$

- For each stationary point find the value of $\frac{d^2 y}{dx^2}$ at the stationary point

- i.e. substitute the x-coordinate of the stationary point into $\frac{d^2 y}{dx^2}$ and evaluate



Your notes

- If $\frac{d^2y}{dx^2}$ is **positive** then the point is a **local minimum**
- If $\frac{d^2y}{dx^2}$ is **negative** then the point is a **local maximum**
- If $\frac{d^2y}{dx^2}$ is **zero** then the point could be a **local minimum**, a **local maximum** OR a **point of inflection**
 - In this case you will need to use method A instead



Examiner Tips and Tricks

- Usually, using the second derivative (Method B above) is a much quicker way of determining the nature of a stationary point
- Sketching the curve can also help



Worked Example

Find the stationary points of

$$y = x^2(2x^2 - 4)$$

and determine the nature of each.

Start by expanding the brackets in the expression for y

$$y = 2x^4 - 4x^2$$

Step 1 Find the first derivative

$$\frac{dy}{dx} = 8x^3 - 8x$$

Step 2 Solve $\frac{dy}{dx} = 0$

$$8x^3 - 8x = 0$$



Your notes

Divide through by 8

$$x^3 - x = 0$$

Fully factorise, spotting the difference of two squares

$$\begin{aligned}x(x^2 - 1) &= 0 \\x(x - 1)(x + 1) &= 0\end{aligned}$$

Solve to find the x -coordinates of the stationary (turning) points

$$x_1 = 0, x_2 = 1, x_3 = -1$$

Step 3 Find the corresponding y -coordinates

$$\begin{aligned}x_1 = 0, \quad y_1 &= (0)^2(2(0)^2 - 4) = 0 \\x_2 = 1, \quad y_2 &= (1)^2(2(1)^2 - 4) = -2 \\x_3 = -1, \quad y_3 &= (-1)^2(2(-1)^2 - 4) = -2\end{aligned}$$

Write the answers as coordinates, being careful to correctly match x 's and y 's

The stationary points are (0, 0), (1, -2) and (-1, -2)

To find the nature of each stationary point, use Method B Step 4 Find the second derivative

$$\frac{d^2y}{dx^2} = 24x^2 - 8$$

Step 5 Evaluate the second derivative at each stationary point

$$\begin{aligned}x = 0, \quad \frac{d^2y}{dx^2} &= 24(0)^2 - 8 = -8 \quad (< 0) \\x = 1, \quad \frac{d^2y}{dx^2} &= 24(1)^2 - 8 = 16 \quad (> 0) \\x = -1, \quad \frac{d^2y}{dx^2} &= 24(-1)^2 - 8 = 16 \quad (> 0)\end{aligned}$$

The nature of each stationary point is now determined

$(0, 0)$ is a local minimum point
 $(1, -2)$ is a local minimum point
 $(-1, -2)$ is a local minimum point



Your notes



Your notes

Modelling with Differentiation

Modelling with Differentiation

How is differentiation used in modelling questions?

- **Derivatives** can be calculated for any variables – not just y and x
- The derivative is a formula giving the **rate of change** of one variable with **respect** to the other variable
 - For example if $A = 4\pi r^2$ then $\frac{dA}{dr} = 8\pi r$
 - $\frac{dA}{dr}$ is the **rate of change** of A with **respect** to r
- The phrase 'increasing at a rate of' means the **rate of change** of one variable with **respect** to time
 - $\frac{d}{dt}$
- **Differentiation** can be used to find **maximum** and **minimum** points of a function
 - In modelling, this is called **optimisation**
 - Second derivative tests help to determine if the point is a maximum or minimum



Examiner Tips and Tricks

- Read the question carefully to determine which variables you will need to use
 - The question may give you a formula to help you



Worked Example

The volume, V , of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.

Find the rate of change of the volume with respect to the radius.



Your notes

Differentiate the formula given for the volume of a sphere.

$$\frac{dV}{dr} = 3 \times \frac{4}{3} \pi \times r^2$$

$$\frac{dV}{dr} = 4\pi r^2$$

Optimisation

What is optimisation?

- In general, optimisation is finding the best way to do something
- In mathematics, optimisation is finding the maximum or minimum output of a function
 - For example, finding the maximum possible profit or minimum costs
- Differentiation can be used to **solve optimisation problems** in **modelling** questions
 - For example you may want to
 - **Maximise** the **volume** of a **container**
 - **Minimise** the **amount** of **fuel** used



Your notes

e.g. AT TIME t SECONDS THE HEIGHT OF A BALL ABOVE THE GROUND IN METRES IS GIVEN BY FORMULA

$$h = 20t - 5t^2$$

USE DIFFERENTIATION TO FIND THE MAXIMUM HEIGHT REACHED BY THE BALL.

$$\frac{dh}{dt} = 20 - 10t$$

$$20 - 10t = 0$$

$$t = 2$$

THE GRAPH OF $20t - 5t^2$ IS A "DOWN" PARABOLA. THE MAXIMUM VALUE OCCURS WHEN $\frac{dh}{dt} = 0$

WHEN $t = 2$,

$$h = 20(2) - 5(2)^2 = 20 \text{ METRES}$$

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Examiner Tips and Tricks

- Exam questions on this topic will often be divided into two parts:
 - First a 'Show that...' part where you derive a given formula from the information in the question
 - And then a 'Find...' part where you use differentiation to answer a question about the formula
- Even if you can't answer the first part you can still use the formula to answer the second part



Worked Example



Your notes

A cuboid has length $4x$ cm, width x cm, and height $\left(\frac{3}{x} - 5\right)$ cm.

(a) Show that the volume, V cm³ is given by $V = 12x - 20x^2$.

The volume of a cuboid is " $V = \text{length} \times \text{width} \times \text{height}$ "

$$\therefore V = 4x \times x \times \left(\frac{3}{x} - 5\right)$$

Expand and simplify

$$V = 4x^2 \left(\frac{3}{x} - 5\right)$$
$$V = \frac{12x^2}{x} - 20x^2$$

$$\therefore V = 12x - 20x^2$$

(b) Find the maximum volume of the cuboid. Differentiate V with respect to x

$$\frac{dV}{dx} = 12 - 40x$$

At the maximum volume, $\frac{dV}{dx} = 0$

$$\therefore 12 - 40x = 0$$

Solve for x

$$40x = 12$$
$$x = \frac{12}{40} = 0.3$$

So the value of x , at the maximum volume is 0.3 Find the maximum volume by substituting $x = 0.3$ in to the formula for V

$$V = 12(0.3) - 20(0.3)^2 = 1.8$$

The maximum volume of the cuboid is 1.8 cm³



Your notes

(c) Prove that your answer is a maximum value.

Using the second derivative is usually the easiest way to find the nature of a stationary point

$$\frac{d^2y}{dx^2} = -40 \quad (< 0)$$

The value of the second derivative (at $x = 0.3$) is negative

Therefore $V = 1.8 \text{ cm}^3$ is a maximum volume



Your notes

Connected Rates of Change

Connected Rates of Change

What are connected rates of change?

- In situations involving more than two variables you can use the **chain rule** to connect multiple rates of change into a single equation
- Look out for the word "**rate**"
 - This implies a **derivative** is involved

- e.g. The height is **increasing** at a **rate of 4 cm per second** means $\frac{dh}{dt} = 4$

- If the variable is **decreasing** then the rate will be **negative**

- e.g. The height is **decreasing** at a rate of 4 cm per second means $\frac{dh}{dt} = -4$



Your notes

e.g. THE VOLUME V , OF A CUBE IS INCREASING AT A CONSTANT RATE OF 3 cm^3 PER SECOND. ASSUMING THAT THE SHAPE REMAINS CUBICAL AT ALL TIMES, FIND THE RATE OF CHANGE OF THE SIDE LENGTH, s , IN cm PER SECOND AT THE MOMENT WHEN $s = 5 \text{ cm}$.

$$\frac{dV}{dt} = 3 \text{ cm}^3/\text{sec}$$

WE KNOW THIS

WE WANT TO KNOW THIS

BY THE CHAIN RULE $\frac{dV}{dt} = \frac{dV}{ds} \times \frac{ds}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

FOR A CUBE, $V = s^3$

SO $\frac{dV}{ds} = 3s^2$

USE DIFFERENTIATION TO GET THE 'MISSING' DERIVATIVE FROM THE CHAIN RULE EQUATION

SO $\frac{ds}{dt} = \frac{\frac{dV}{dt}}{\frac{dV}{ds}} = \frac{3}{3s^2} = \frac{1}{s^2}$

REARRANGE

WHEN $s = 5$,

$\frac{ds}{dt} = \frac{1}{5^2} = \frac{1}{25} \text{ cm/sec}$

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Examiner Tips and Tricks

- These problems can involve a lot of letters

- be sure to keep track of what they all refer to
- Be especially sure that you are clear about which letters are **variables** and which are **constants**
 - these behave very differently when differentiation is involved!



Worked Example



Your notes



Your notes

? In a manufacturing process, plastic spheres are produced in such a way that the volume, V , of a sphere increases at a constant rate of 10 cm^3 per second. Find the rate of change of the surface area, A , of a sphere at the moment when the surface area is equal to $32\pi \text{ cm}^2$.

$$\frac{dV}{dt} = 10 \text{ cm}^3/\text{sec}$$

WE KNOW THIS

WE WANT TO KNOW THIS

BY THE CHAIN RULE $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

$$V = \frac{4}{3}\pi r^3 \quad (\text{WHERE } r \text{ IS THE RADIUS})$$

$$A = 4\pi r^2$$

$$A^{\frac{3}{2}} = (4\pi r^2)^{\frac{3}{2}} = 8\pi^{\frac{3}{2}} r^3 = 8\pi\sqrt{\pi} r^3$$

$$\frac{8\pi\sqrt{\pi} r^3}{6\sqrt{\pi}} = \frac{4}{3}\pi r^3 \quad \text{SO} \quad V = \frac{1}{6\sqrt{\pi}} A^{\frac{3}{2}}$$

$$\text{AND } \frac{dV}{dA} = \frac{3}{2} \times \frac{1}{6\sqrt{\pi}} A^{\frac{3}{2}-1} = \frac{1}{4\sqrt{\pi}} A^{\frac{1}{2}}$$

WRITE V IN TERMS OF A

DIFFERENTIATE TO FIND THE 'MISSING' DERIVATIVE



Your notes

$$\text{SO } \frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$$

$$10 = \frac{1}{4\sqrt{\pi}} A^{\frac{1}{2}} \left(\frac{dA}{dt} \right)$$

$$\text{AND } \frac{dA}{dt} = \frac{40\sqrt{\pi}}{A^{\frac{1}{2}}} = 40\sqrt{\frac{\pi}{A}}$$

$$\text{WHEN } A = 32\pi,$$

$$\frac{dA}{dt} = 40\sqrt{\frac{\pi}{32\pi}} = \frac{40}{\sqrt{32}} = 5\sqrt{2} \text{ cm}^2/\text{sec}$$

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SUBSTITUTE
AND REARRANGE